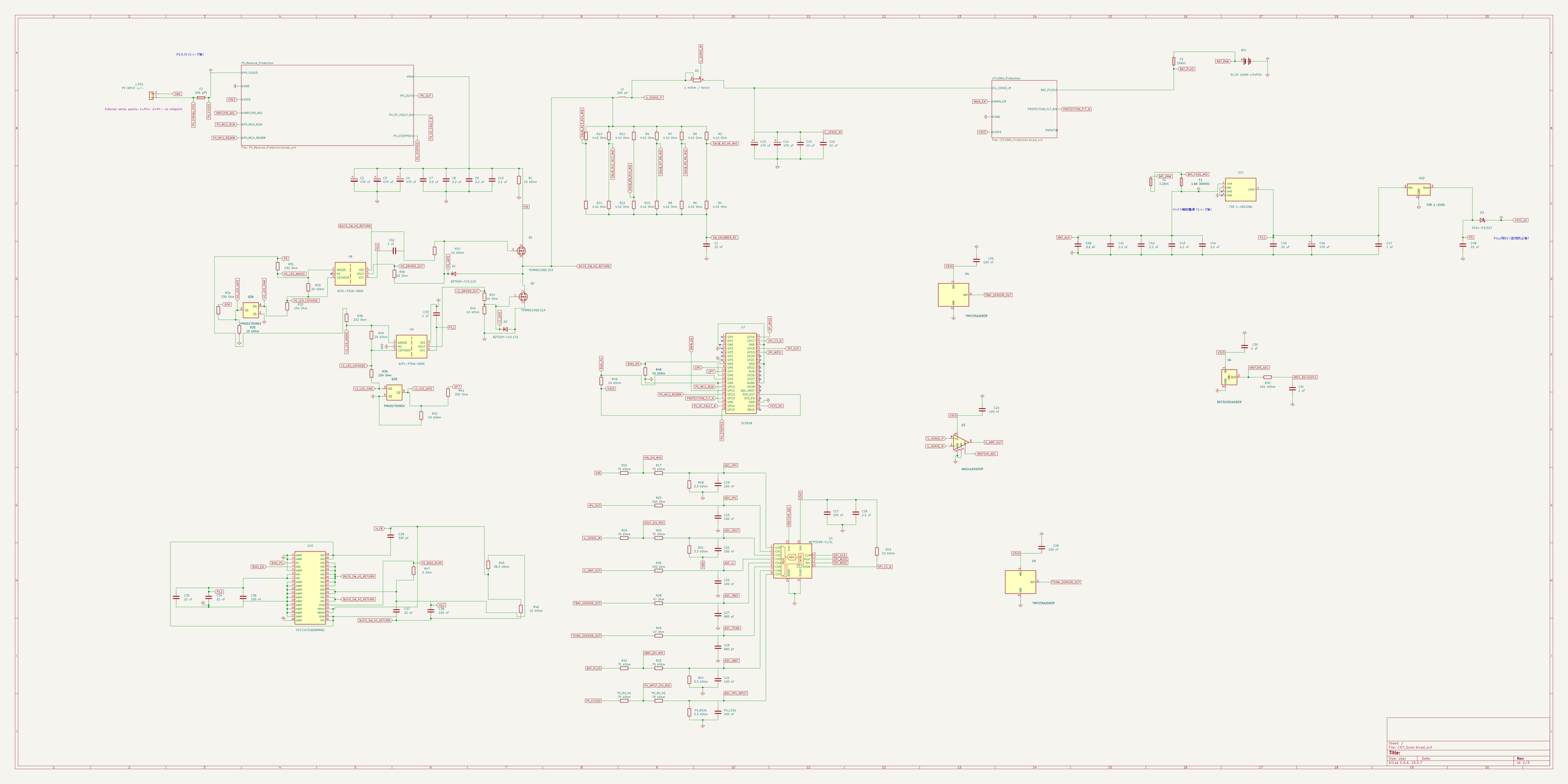
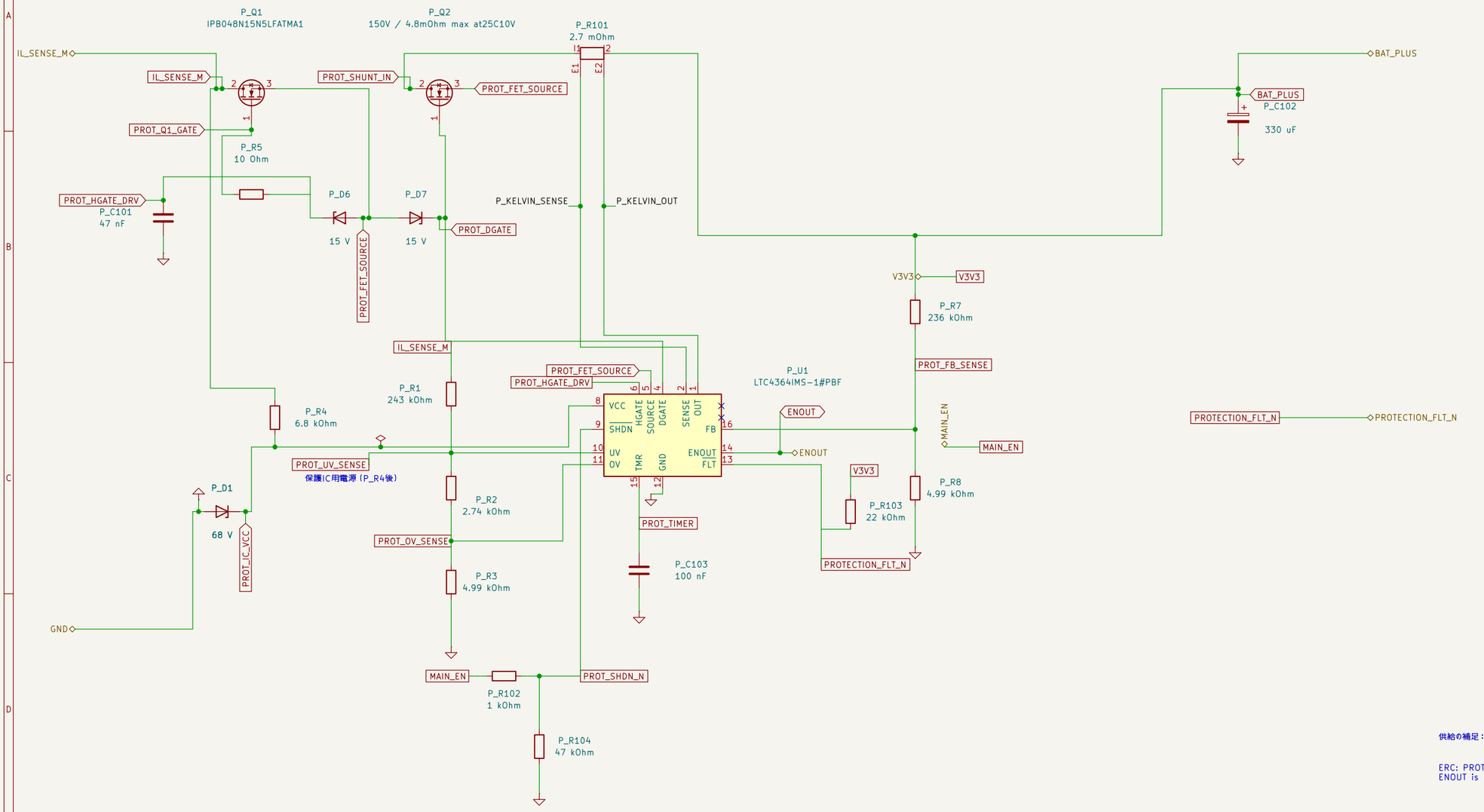

PV入力（ヒューズ後）
J_PV1
PV INPUT +/-
External series panels: 1=PV+; 2=PV-; no midpoint
GND
F2
20A gPV
PV_FUSED
PV_Reverse_Protection
PV_FUSED
GND
3V3
VREF2V5_ADC
PV_MCU_RUN
PV_MCU_REARM
VIN
IPV_OUT
PV_DC_FAULT_N
PV_STOPPED
PV_Reverse_Protection.kicad_sch
PV_STOPPED
PV_DC_FAULT_N
C2 470 uF
C3 470 uF
C4 470 uF
C7 3.3 uF
C8 2.2 uF
C9 2.2 uF
C10 2.2 uF
R1 15 kOhm
VIN
IL_SENSE_M
R2
1 mOhm / Kelvin
IL_SENSE_P
L1
200 uF
SW_SNUBBER_RC
R13 4.42 Ohm
R11 4.42 Ohm
R9 4.42 Ohm
R7 4.42 Ohm
R5 4.42 Ohm
R3 4.42 Ohm
R14 4.42 Ohm
R12 4.42 Ohm
R10 4.42 Ohm
R8 4.42 Ohm
R6 4.42 Ohm
R4 4.42 Ohm
SNUB_R3_R5_MID
C13 470 uF
C14 470 uF
C15 10 uF
C16 10 uF
C1 22 nF
LTC4364_Protection
DIL_SENSE_M
MAIN_EN
GND
3V3
BAT_PLUS
PROTECTION_FLT_N
ENOUT
LTC4364_Protection.kicad_sch
F3
25ADC
BAT_RAW
BAT_PLUS
BT1
51.2V 100Ah LiFePO4
BUCK_SW_HS_RETURN
C52 1 uF
R37 10 kOhm
Q1
TK9R6E15QS,S1X
HS_DRIVER_OUT
HS_GATE
D1
BZT52H-C15,115
R36 10 Ohm
BUCK_SW_HS_RETURN
LS_DRIVER_OUT
R43 10 Ohm
R44 10 kOhm
Q2
TK9R6E15QS,S1X
LS_GATE
D2
BZT52H-C15,115
C53 1 uF
F12
U8
ACPL-P346-560E
U9
ACPL-P346-560E
F5
R31 232 Ohm
HS_LED_ANODE
R33 20 kOhm
HS_LED_CATHODE
R32 154 Ohm
R34 330 Ohm
HS_GATE_DRIVE
HS_GATE
Q3A
PMGD175XNEX
R35 10 kOhm
R38 232 Ohm
LS_LED_ANODE
R40 20 kOhm
LS_LED_CATHODE
R39 154 Ohm
Q3B
PMGD175XNEX
LS_LED_SINK
LS_LED_GATE
R41 330 Ohm
R42 10 kOhm
GP7
U7
RP2040
SPI_CS_N
SPI_SCK
SPI_MOSI
VSYS_5V
SC0918
BIAS_EN
R48 10 kOhm
R49 10 kOhm
PV_MCU_REARM
PROTECTION_FLT_N
PV_DC_FAULT_N
PV_STOPPED
U4
TMP235A20BZR
C26 100 nF
TBAT_SENSOR_OUT
U3
INA241A3IDDFR
C24 100 nF
IL_AMP_OUT
VREF2V5_ADC
U6
REF5025EA/DBZR
C30 1 uF
VREF_DECOUPLE
R50 100 mOhm
C31 1 uF
U5
TMP235A20BZR
C28 100 nF
TSINK_SENSOR_OUT
U10
UCC14151QDWRRQ1
HI_FB
C39 330 pF
R45 38.3 kOhm
R46 10 kOhm
R47 0 Ohm
HS_BIAS_RLIM
BIAS_PG
BIAS_EN
BUCK_SW_HS_RETURN
C35 22 uF
C36 100 nF
C37 22 uF
C38 100 nF
U1
MCP3208-CI/SL
R15 10 kOhm
SPI_CS_N
SPI_SCK
SPI_MOSI
SPI_MISO
VREF2V5_ADC
C17 100 nF
C18 2.2 uF
VIN_DIV_MID
R16 75 kOhm
R17 75 kOhm
ADC_VPV
R18 3.3 kOhm
C19 100 nF
IPV_OUT
R25 100 Ohm
ADC_IPV
C23 100 nF
VOUT_DIV_MID
IL_SENSE_M
R19 75 kOhm
R20 75 kOhm
ADC_VOUT
R21 3.3 kOhm
C20 100 nF
IL_AMP_OUT
R26 100 Ohm
ADC_IL
C25 100 nF
TBAT_SENSOR_OUT
R28 47 Ohm
ADC_TBAT
C27 680 pF
TSINK_SENSOR_OUT
R29 47 Ohm
ADC_TSINK
C29 680 pF
VBAT_DIV_MID
BAT_PLUS
R22 75 kOhm
R23 75 kOhm
ADC_VBAT
R24 3.3 kOhm
C21 100 nF
PV_INPUT_DIV_MID
PV_FUSED
PV_RV_H1 75 kOhm
PV_RV_H2 75 kOhm
ADC_VPV_INPUT
PV_RSS 3.3 kOhm
PV_C101 100 nF
BAT_RAW
F4
3.24DC
BAT_FUSE_MID
F6
1.6A 300VDC
U11
TSR 1-48120WI
バッテリ補助電源（ヒューズ後）
U12
TSR 1-2450
BAT_AUX
C40 2.2 uF
C41 2.2 uF
C42 2.2 uF
C43 2.2 uF
C44 2.2 uF
F22
C45 22 uF
C46 470 uF
C47 1 uF
D3
SS14-E3/61T
VSYS_5V
F5
C48 22 uF
Pico用5V（逆流防止後）
Sheet: /
File: C07_Solar.kicad_sch
Title:
Size: User
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 1/3

WSL3637: I1/I2 = current; E1/E2 = dedicated Kelvin sense.
Do not merge sense traces into the load-current copper.
I/E names follow datasheet p2; footprint mapping is pending.



I旧原本についての説明（下記）／現図は4端子へ修正済み
現在のP_R101：I1・I2は主電流、E1・E2は専用検出線。
フットプリントは未確定。ENOUTは未使用。

SOURCE MODEL VIEW — WSL3637 (P_R101) is a two-terminal representation.
Physical part has four terminals; Kelvin wiring and footprint remain unqualified.
ENOUT is left open in the source model.

供給の補足：旗はP_R4後の配線上へ移設済み

ERC: PROT_IC_VCC supplied from VIN via P_R4.
ENOUT is unused (LTC4364 datasheet p9).

Sheet: /LTC4364_Protection/	
File: LTC4364_Protection.kicad_sch	
Title:	
Size: A3	Date:
KiCad E.D.A. 10.0.7	**Rev:** Id: 2/3

PV_FUSED
R27
IP
KP
KM
IM
1 mOhm / Kelvin
PV_AFTER_SHUNT
PV_D102
K
V20PW22-M3/I
L2
62 uH
PV_L2_OUT
PV_Q1
PV_FET_SOURCE
IPB048N15N5LFATMA1
PV_Q2
PV_FET_GATE
IPB048N15N5LFATMA1
VIN
PV_D104
V20PW22-M3/I
PV_D105
V20PW22-M3/I
PV_GATE_RC
PV_R118
10 Ohm
PV_C107
100 nF
PV_D103
BZT52H-C15,115
PV_FUSED
PV_R115
330 kOhm
PV_DRIVER_RUN
PV_R116
10.2 kOhm
PV_C106
100 nF
PV_FUSED
PV_VCCUV
PV_DRIVER_VCC
PV_C105
1 uF
GND
PV_R117
113 kOhm
GND
V3V3
PV_R104
22 kOhm
PV_OC_FAULT_N
PV_U1
RUN
VIN
VCC
VCCUV
FAULT_N
TIMER
INP
OVLO
GND
SNS+
SNS-
BST
TS
TGUP
TGDN
IMON
ISET
LTC7000ARMSE#PBF
PV_SNS_MINUS
PV_R103
100 Ohm
PV_AFTER_SHUNT
PV_C102
10 uF
PV_D101
PV_BOOTSTRAP
PV_DRIVER_VCC
MMBD1501A
PV_R105
100 kOhm
PV_DRIVER_TGUP
PV_R106
10 MOhm
PV_DRIVER_TIMER
PV_R114
100 kOhm
PV_C104
1 nF
PV_OC_FAULT_N
PV_OC_FAULT_NQ
V3V3
C22
100 nF
GND
U2
PV_FUSED
PV_AFTER_SHUNT
INA241A3IDDFR
IPV_OUT
VREF2V5_ADC
PV_FUSED
PV_R108
820 kOhm
PV_DRIVER_OVLO
PV_R109
10 kOhm
V3V3
PV_C_PV_U3
100 nF
PV_LATCH_Q
PV_R107
100 kOhm
PV_U3
VCC
IN+
OUT
IN-
GND
PV_CMP_OUT
TLV3201AIDBVR
PV_R111
10.2 kOhm
PV_CMP_REF
PV_R112
9.8 kOhm
PV_C103
10 nF
PV_MCU_RUN
PV_R113
100 kOhm
PV_U4
VCC
GND
V3V3
PV_C_PV_U4
100 nF
PV_LATCH_CLR_N
PV_SUP_RESET_N
SN74LVC1G11DBVR
PV_R102
100 kOhm
PV_MCU_REARM
PV_R110
100 kOhm
PV_U5
CLK
D
Q
GND
VCC
PRE
CLR
Q
V3V3
PV_STOPPED
SN74LVC1G74DCTR
PV_LATCH_Q
V3V3
PV_C_PV_U5
100 nF
V3V3
PV_C108
100 nF
PV_V3V3_SENSE
PV_R_SUP31
31.2 kOhm
V3V3
PV_R_SUP32
10 kOhm
PV_R119
10 kOhm
PV_U6
VDD
SENSE1
SENSE2
GND
RESET1_N
RESET2_N
NC
NC
TPS37042BJOFDDFRQ1
PV_R_SUP25
21.3 kOhm
VREF2V5_ADC
PV_R_SUP26
10 kOhm
PV_REF2V5_SENSE
ERC: PV_FUSED supplied from PV strings via F2.
供給元：親シートのPV入力とF2（旗も同位置）
Sheet: /PV_Reverse_Protection/
File: PV_Reverse_Protection.kicad_sch
Title:
Size: A2
Date:
Rev:
KiCad E.D.A. 10.0.7
Id: 3/3